

Statistical Modeling I

HES 505 Fall 2025: Session 23

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Objectives

By the end of today you should be able to:

- **Define** the likelihood function and its relationship to statistical inference.
- **Recognize** key assumptions of statistical models and how spatial data may challenge those assumptions.
- **Simulate** fake data with known relationships
- **Fit** simple linear and generalized linear models to spatial data

Inference for First Order Properties

$$z(\mathbf{x}) = \mu(\mathbf{x}) + \epsilon(\mathbf{x})$$

- Often we actually care about the “drivers” of $\mu(\mathbf{x})$
- Inference not prediction
- “Which spatial attributes drive $z(\mathbf{x})$?”
- Does $\mu(\mathbf{x})$ increase or decrease with changes in a particular variable?”

Using regression to estimate μ

$$z(\mathbf{s}) \sim \text{Distr}(\mu, \sigma) \\ \mu = w_0 + \sum_{i=1}^m w_i X_i(\mathbf{s}) + \epsilon$$

- When $z(\mathbf{s})$ is binary $\rightarrow$ logistic regression
- When $z(\mathbf{s})$ is continuous $\rightarrow$ linear (gamma) regression
- When $z(\mathbf{s})$ is discrete $\rightarrow$ Poisson regression
- Assumptions about (ϵ) matter!!

Common Regression Forms

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\[ \begin{equation} y \sim N(\bar{y}, \bar{\sigma}) \quad y = \\ w_0 + \sum_{i=1}^m w_i X_i(\mathbf{s}) + \epsilon \\ \newline y \sim \text{Bern}(p) \quad \text{logit}(y) = w_0 + \\ \sum_{i=1}^m w_i X_i(\mathbf{s}) + \epsilon \\ \newline \end{equation} y \sim \text{Poisson}(\lambda) \quad \\ \log(\lambda) = w_0 + \sum_{i=1}^m w_i X_i(\mathbf{s}) \\ + \epsilon \quad \]
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Key components

- **Distributional** assumptions - the likelihood
- (w_i) is 'spatial weight', equivalent to (β) in typical regression
- *link* function scales linear model to appropriate support

Estimating parameters

- You measure $\langle y \rangle$
- Your model expresses your hypothesis about the rules governing $\langle y \rangle$
- We need estimates of $\langle w_i \rangle$ to complete our rule

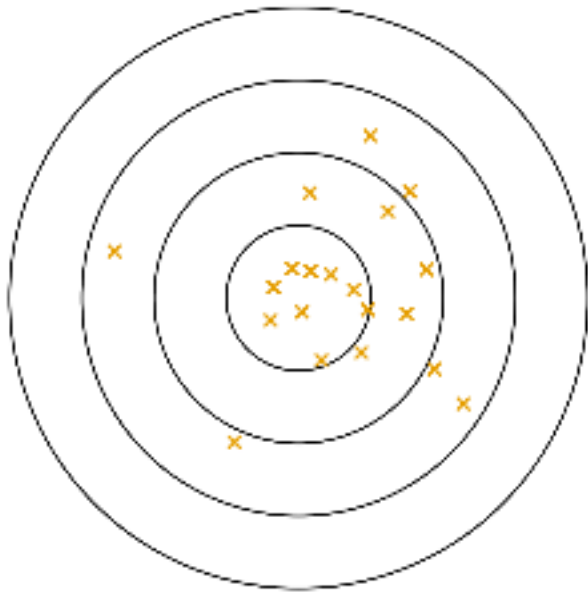
Estimating parameters via the Likelihood Function

$$\begin{aligned} p(y_i \mid \beta, \sigma) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \mathbf{w}_i^T \boldsymbol{\beta})^2}{2\sigma^2}\right) \\ p(\mathbf{y} \mid \beta, \sigma) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \mathbf{w}_i^T \boldsymbol{\beta})^2}{2\sigma^2}\right) \end{aligned}$$

Estimating parameters via the likelihood function

- Maximum Likelihood: treats $\backslash(w_i\backslash)$ as fixed, unknown constants. Uncertainty calculated after...
- Bayesian = $\backslash(w_i\backslash)$ is a random quantity which varies within the constraints of our *prior*. Uncertainty built into the estimation process.

Model Predictions (Center) vs. Observed Data (Darts)



- MLE: “What’s the best dart throw?”
- Bayesian: “Given where darts tend to land, what is the distribution of possible aiming points?”

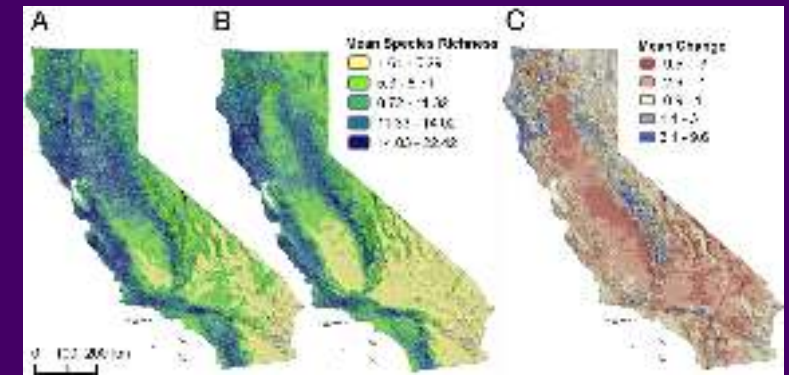
Deriving model assumptions from the likelihood

- Shape of likelihood function gives an indication of residual behavior
- $\prod$ signals independent observations
- Moments of the distribution add assumptions (i.e. Poisson mean = variance)

Logistic Regression and Distribution Models

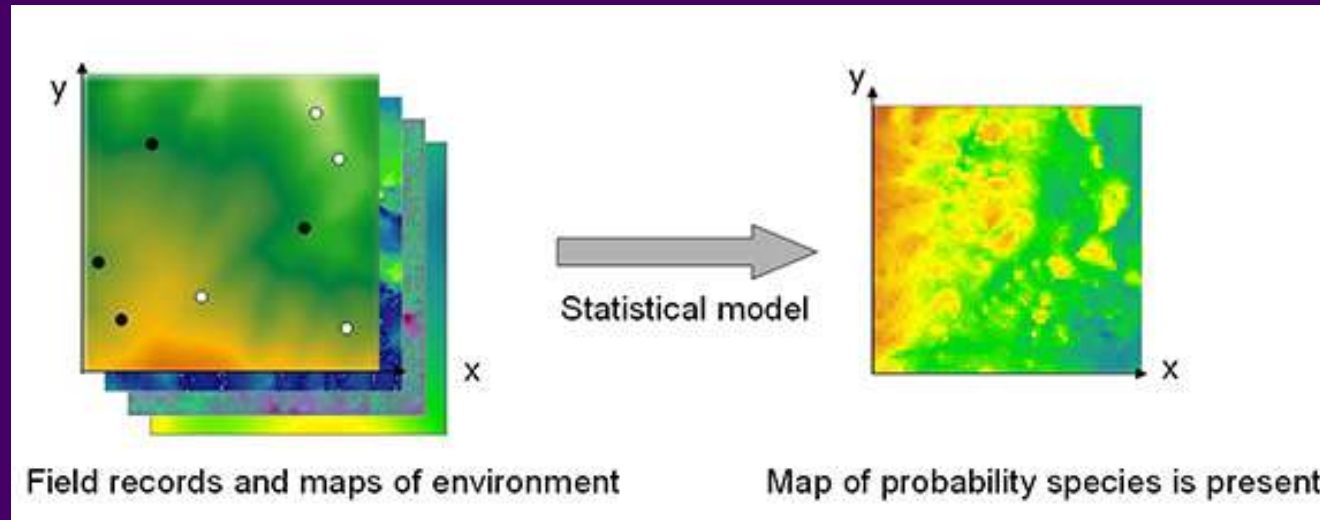
Why do we create distribution models?

- To identify important correlations between predictors and the occurrence of an event
- Generate maps of the 'range' or 'niche' of events
- Understand spatial patterns of event co-occurrence
- Forecast changes in event distributions



From Wiens et al. 2009

General analysis situation



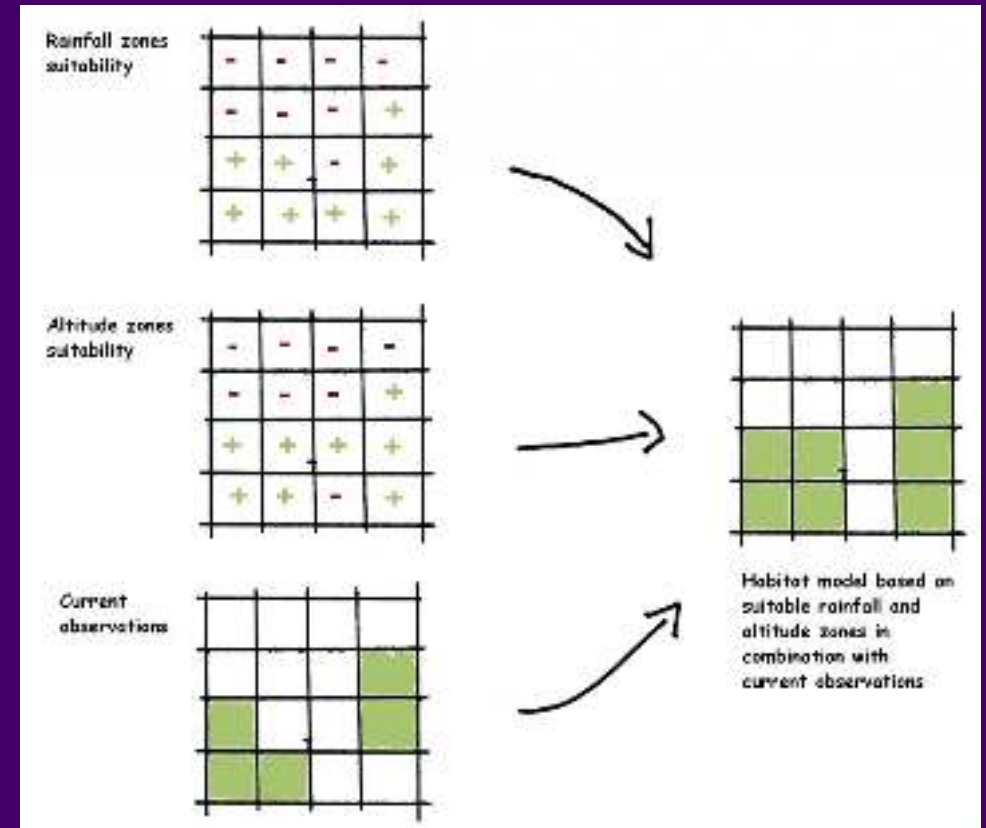
From Long

- Spatially referenced locations of events $((\mathbf{y}))$ sampled from the study extent
- A matrix of predictors $((\mathbf{X}))$ that can be assigned to each event based on spatial location

Goal: Estimate the probability of occurrence of events across unsampled regions of the study area based on correlations with predictors

Modeling Presence-Absence Data

- Random or systematic sample of the study region
- The presence (or absence) of the event is recorded for each point
- Hypothesized predictors of occurrence are measured (or extracted) at each point



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Logistic regression

Linear regression

Logistic regression

Linear regression

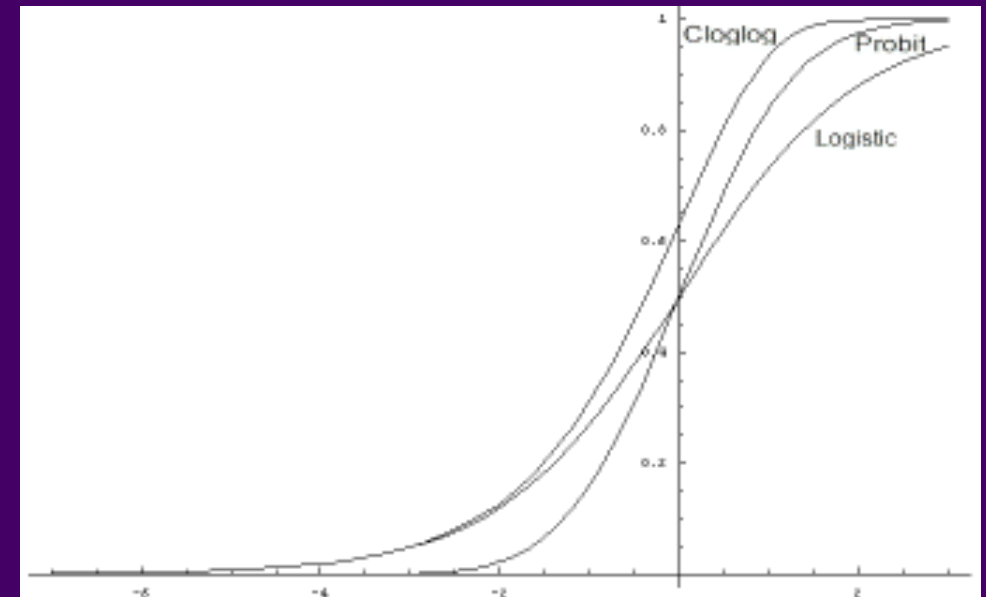
Logistic regression

Linear regression

Logistic regression

- We can model favorability as the **probability** of occurrence using a logistic regression
- A *link* function maps the linear predictor $((\mathbf{x}_i' \beta + \alpha))$ onto the support (0-1) for probabilities
- Estimates of (β) can then be used to generate ‘wall-to-wall’ spatial predictions

$$y_i \sim \text{Bern}(p_i) \\ \text{link}(p_i) = \mathbf{x}_i' \beta + \alpha$$



From Mendoza

Comparison with Machine Learning

- Statistical models describe a generative process: how'd the data get here
- Machine learning models ask something different
- Underlying modes of prediction

